

```

> restart: with(plots): with(plottools):with(LinearAlgebra):with
  (VectorCalculus):BasisFormat(false):with(GE03):
> with(CurveFitting):with(ColorTools):
Farver anvendt i projektet
> mg:=Color([36, 59, 52])
                                 $mg := \langle \text{RGB} : 0.141 \ 0.231 \ 0.204 \rangle$  (1)
> lg:=Color([136, 202, 156])
                                 $lg := \langle \text{RGB} : 0.533 \ 0.792 \ 0.612 \rangle$  (2)

```

1. Konstanter der kan parametriseres

Konstant, der bestemmer hvor meget knobet forskydes i y-retningen:

```

> a:=0.2: 'a'=a
                                 $a = 0.2$  (1.1)

```

Konstant, der angiver knobets bredde i x-retningen:

```

> b:=0.9: 'b'=b
                                 $b = 0.9$  (1.2)

```

Konstant, der angiver knobets bredde i y-retningen:

```

> c:=0.5: 'c'=c
                                 $c = 0.5$  (1.3)

```

Konstant, der bestemmer forskydningen af den spejlvendte version af yx-planet:

```

> d:=2: 'd'=d
                                 $d = 2$  (1.4)

```

Knobets længde:

```

> e:= 0.3: 'e'=e
                                 $e = 0.3$  (1.5)

```

2. Funktion for z koordinatet af råbsknobsfunktionen

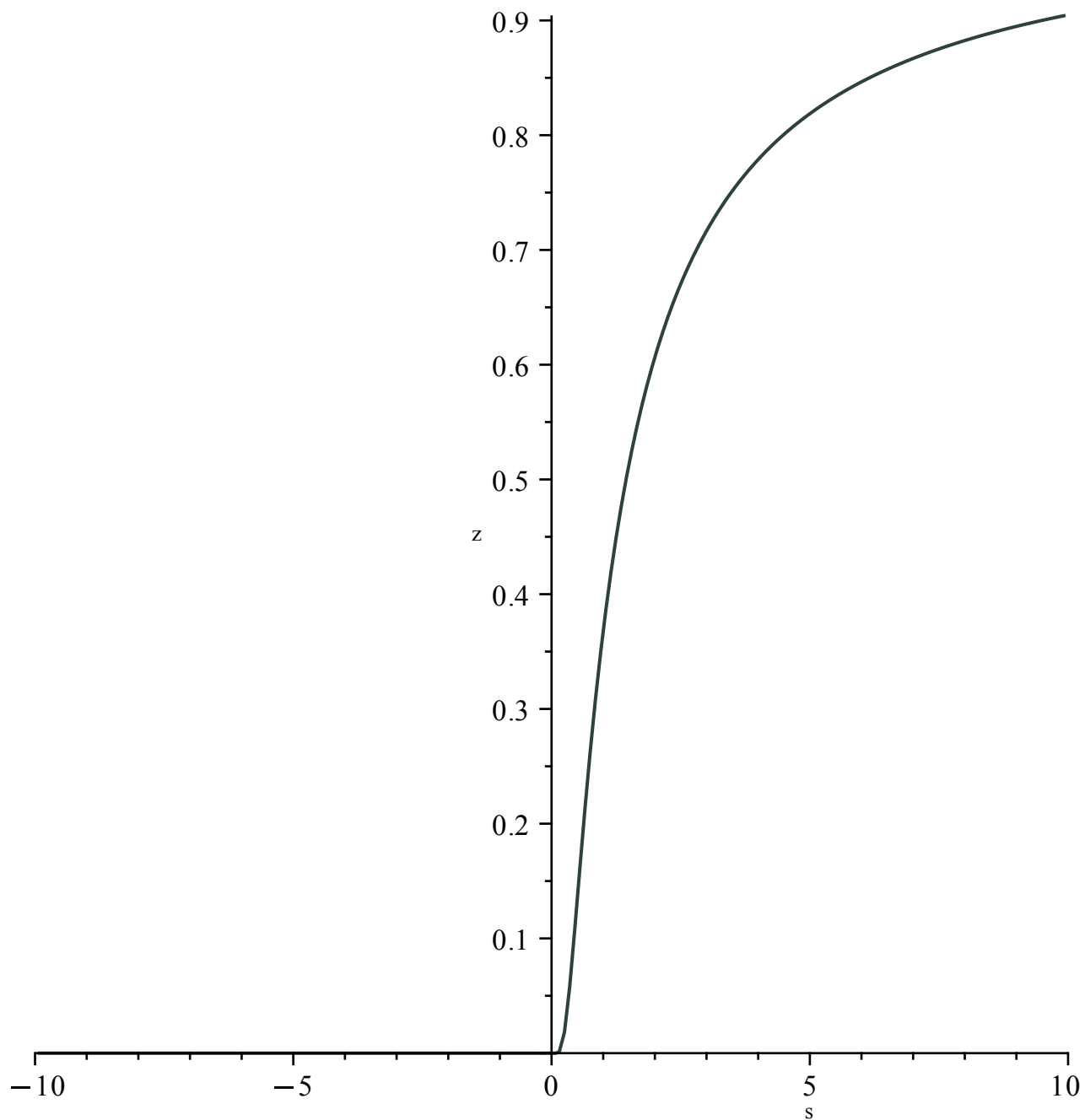
Der anvendes en bump-funktion til at bestemme kurveforløbet for z-koordinaten. Funktionen giver en smooth transition.

```

> f1:= s -> piecewise(s<=0,0,s>0,exp(-1/s)): 'f1(s)'=f1(s)
                                
$$f1(s) = \begin{cases} 0 & s \leq 0 \\ e^{-\frac{1}{s}} & 0 < s \end{cases} \quad (2.1)$$

> plot(f1(s),color=Color([0.136, 0.202, 0.156]),labels=["s","z"]);

```

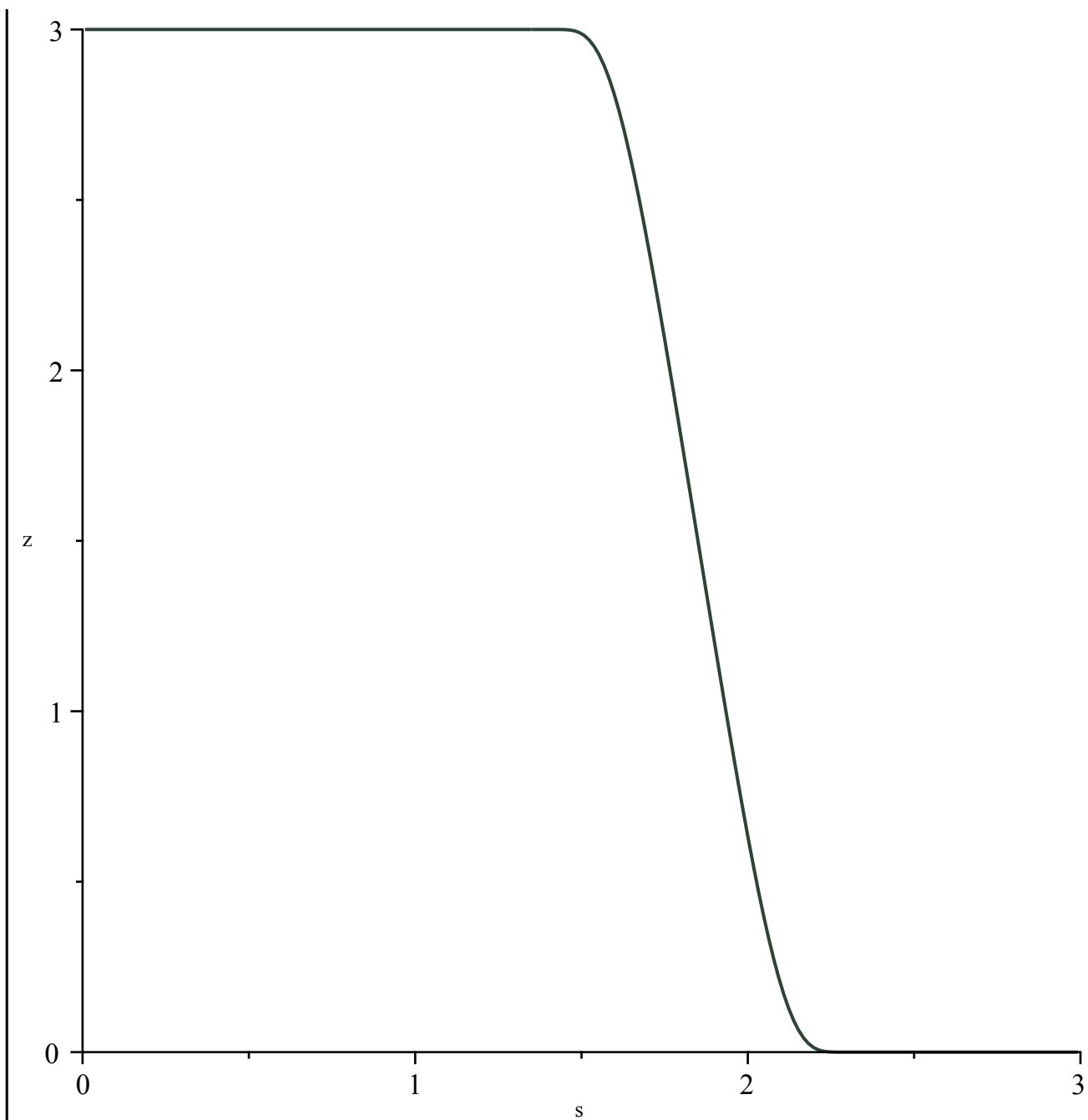


"Torvet" fra højre:

```
> g1:= s -> 3*f1(1 - s + 1.35)/(f1((1 - s) + 1.35) + f1(- 1.35+
s)): 'g1(s)'=simplify(g1(s))
```

$$g1(s) = 3 \left(\begin{cases} 1 & s \leq 1.35 \\ \frac{1}{\frac{1480. - 800. s}{e^{1269. - 1480. s + 400. s^2}} + 1} & s < 2.35 \\ 0 & 2.35 \leq s \end{cases} \right) \quad (2.2)$$

```
> plot(g1(s), s=0..3,color=Color([0.136, 0.202, 0.156]),labels=
["s","z"])
```

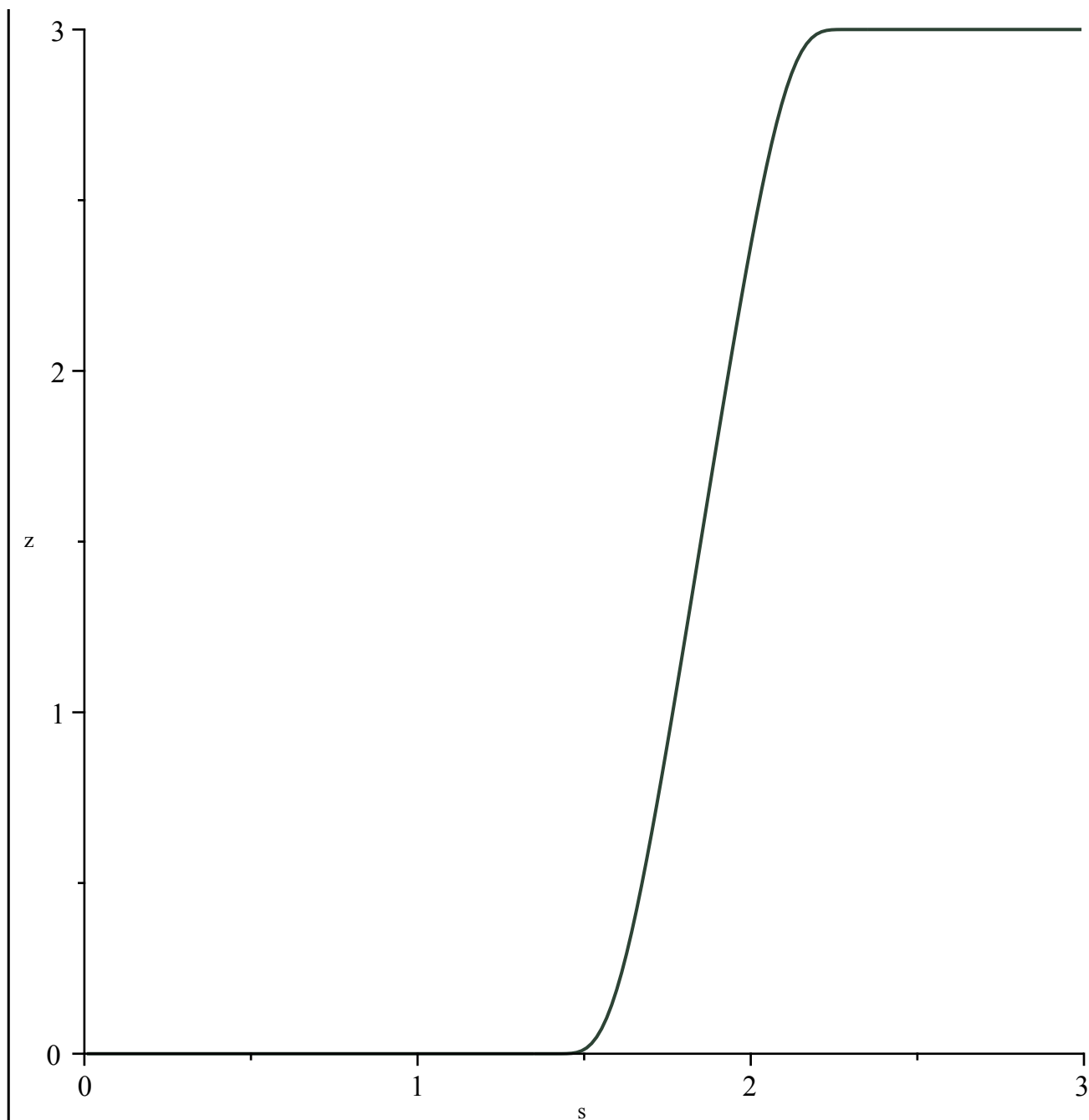


"Torvet" fra venstre:

```
> g2 := s -> 3*f1(s - 1.35)/(f1(s - 1.35) + f1(1 - s - (-1.35)))
: 'g2(s)'=simplify(g2(s))
```

$$g2(s) = 3 \left\{ \begin{array}{ll} 0 & s \leq 1.35 \\ \frac{1}{1 + e^{\frac{-1480. + 800. s}{1269. - 1480. s + 400. s^2}}} & s < 2.35 \\ 1 & 2.35 \leq s \end{array} \right. \quad (2.3)$$

```
> plot(g2(s), s=0..3,color=Color([0.136, 0.202, 0.156]),labels=
["s","z"])
```



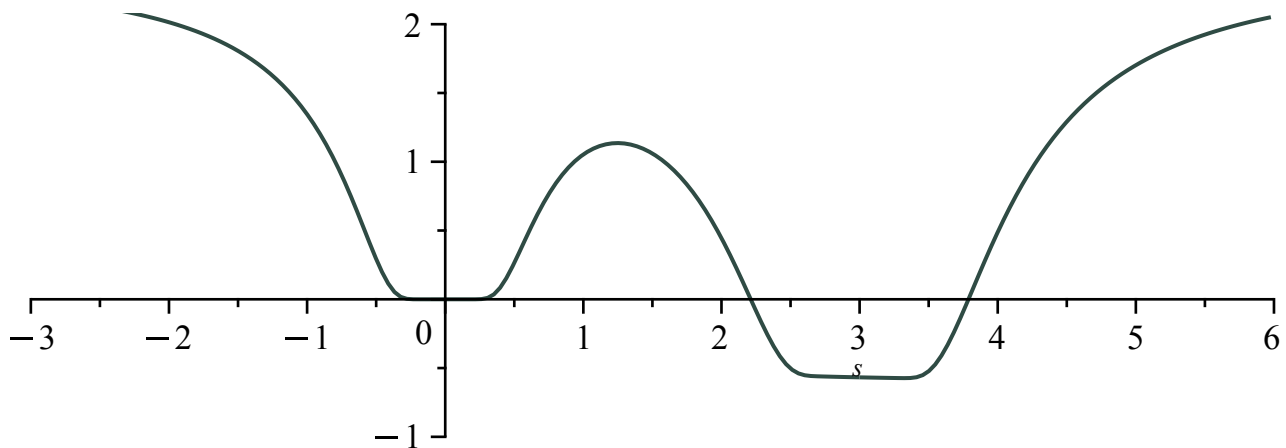
3. Løkkerne

Udtrykket inde i parentesen bestemmer, hvornår og hvordan kurven falder, mens den anden del af funktionen bestemmer, hvornår og hvordan kurven vokser i intervallet $s \in]0; 3[$

```
> f2:= s -> evalf(3*exp(-0.5/s^2)*(-0.2+exp(-1/(3-s)^2))): 'f2(s) '=
f2(s)
```

$$f2(s) = 3 \cdot e^{-\frac{0.5}{s^2}} \left(-0.2 + e^{-\frac{1}{(3-s)^2}} \right) \quad (3.1)$$

```
> plot(f2(s), s=-3...6, scaling=constrained, view=[-3...6, -1...
.2],color=mg);
```

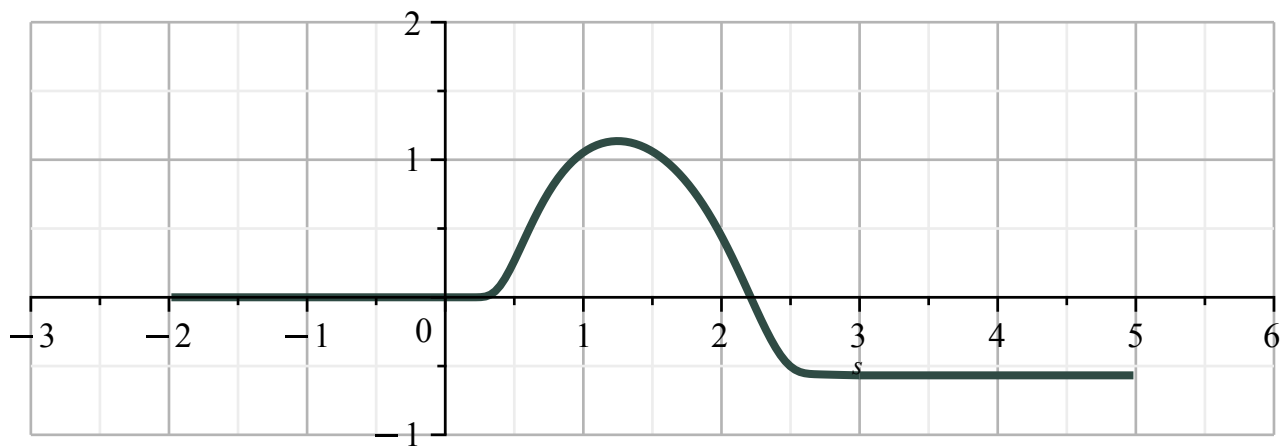


For funktionen gælder, at den kun skal vokse og falde i intervallet $s =]0; 3[$:

```
> f3:= unapply(piecewise(s<0, 0, s>0 and s<3, f2(s), s>3, f2(3.0)), v):'f3(s)'=f3(s);
```

$$f_3(s) = \begin{cases} 0 & s < 0 \\ 3 \cdot e^{-\frac{0.5}{s^2}} \left(-0.2 + e^{-\frac{1.}{(3.-1.s)^2}} \right) & 0 < s < 3 \\ -0.5675756814 & 3 < s \end{cases} \quad (3.2)$$

```
> plot(f3(s), s=-2...5, scaling= constrained, gridlines, color=mg, thickness=3, view=[-3...6, -1...2]);
```

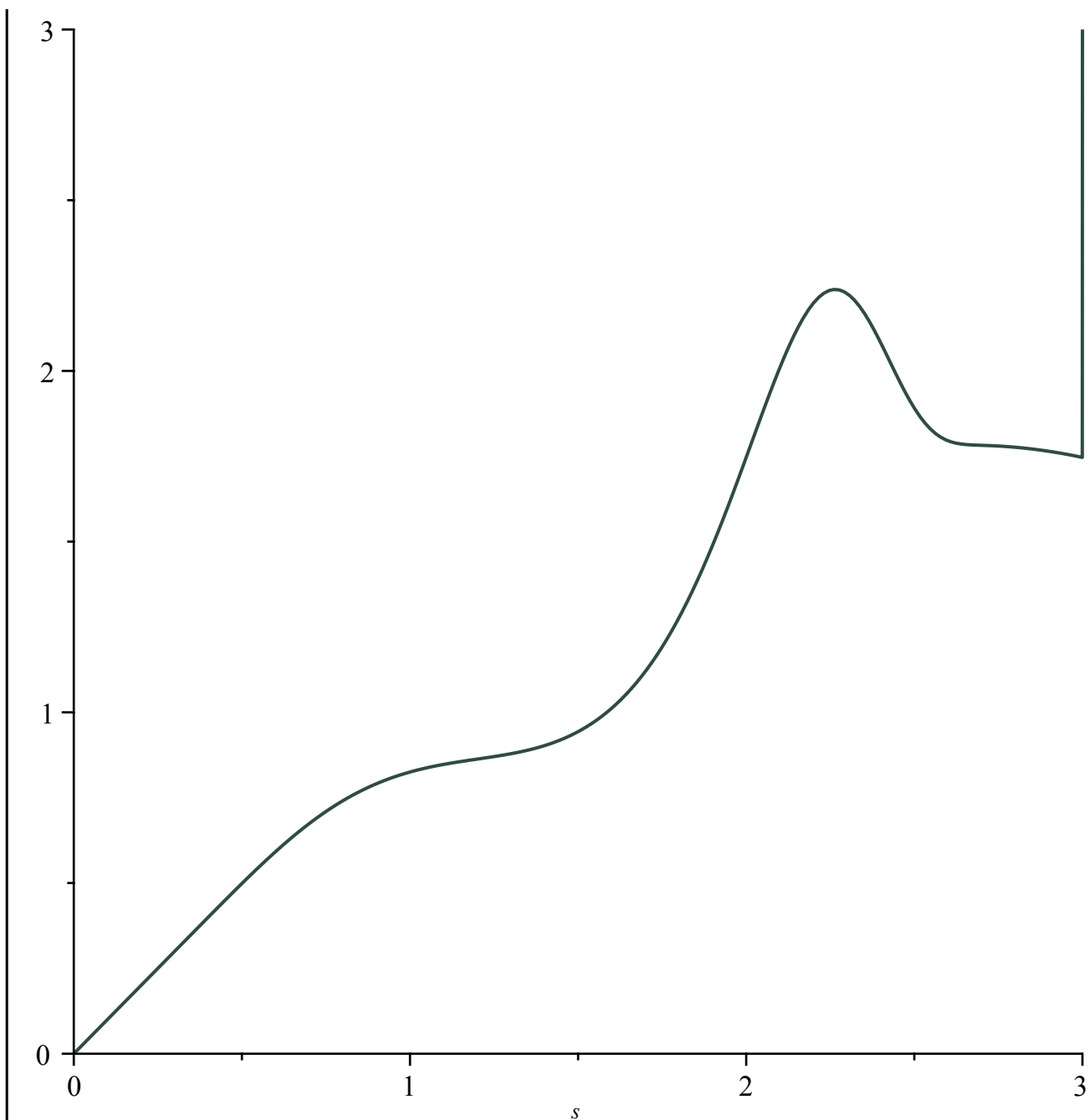


Kurven buelængdeparametriseres

```
> etal:= unapply(evalf(Int(cos(evalf(Int(f3(v), v=0...u))), u=0..s)), s):'etal(s) '=etal(s);
```

$$\eta I(s) = \int_0^s \cos \left(\int_0^u \left(\begin{cases} 0. & s < 0. \\ 3. e^{-\frac{0.5}{s^2}} \left(-0.2 + e^{-\frac{1.}{(3.-1.s)^2}} \right) & 0. < s < 3. \\ -0.5675756814 & 3. < s \\ 0. & otherwise \end{cases} \right) dv \right) du \quad (3.3)$$

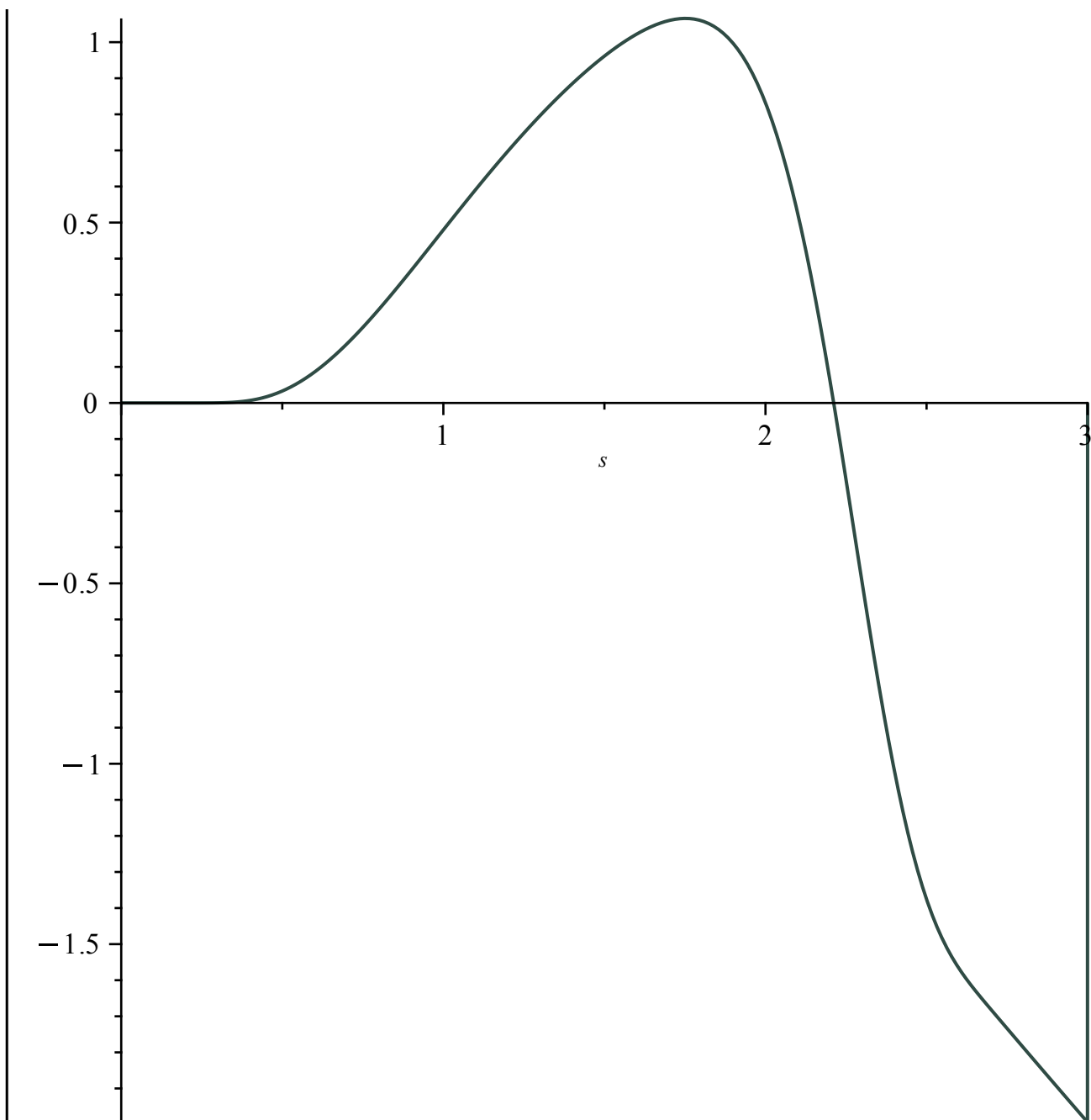
```
> plot(etal(s), s=0...3,color=mg);
```



```
> eta2:= unapply(evalf(Int(sin(evalf(Int(f3(v), v=0...u))), u=0..s)), s);
```

$$\eta_2 := s \mapsto \int_0^s \sin \left(\int_0^u \left(\begin{cases} 0. & s < 0. \\ 3. \cdot e^{-\frac{0.5}{s^2}} \cdot \left(-0.2 + e^{-\frac{1.}{(3.-1.\cdot s)^2}} \right) & 0. < s < 3. \\ -0.5675756814 & 3. < s \\ 0. & otherwise \end{cases} \right) dv \right) du \quad (3.4)$$

```
> plot(eta2(s), s=0...3,color=mg);
```



Intervallerne for løkkerne skal præciseres. Intervallet stopper, når η_2 skærer x-aksen. I knobets endelige funktion ændres η_2 's størrelse med konstanten c og forskydes med a . Dette påvirker skæringen med x-aksen.

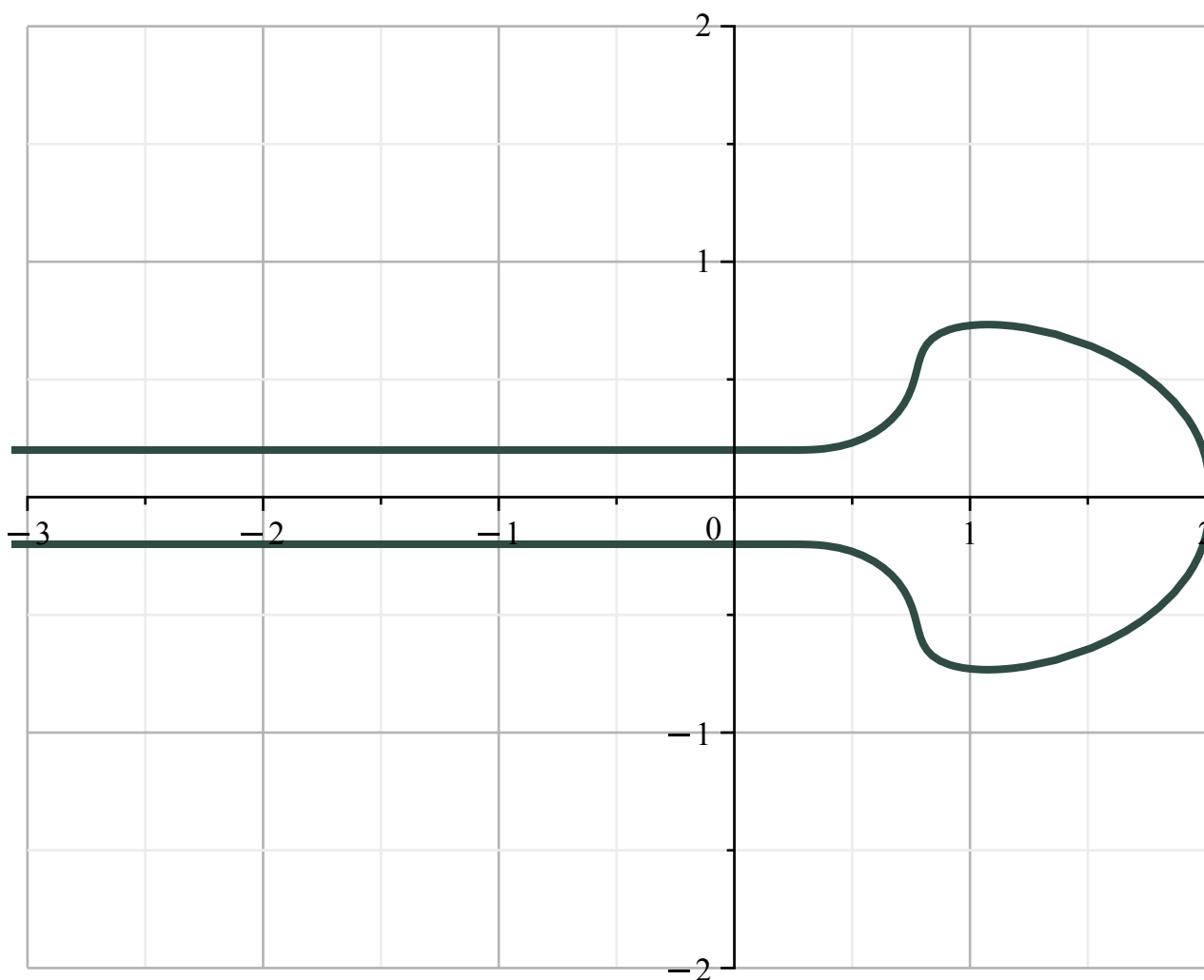
Skæringen bestemmes derfor således:

$$\text{> fsolve}(c \cdot \eta_2(s) + a = 0, s) : kx := \% : 'kx' = kx \quad (3.5)$$

$kx = 2.282839660$

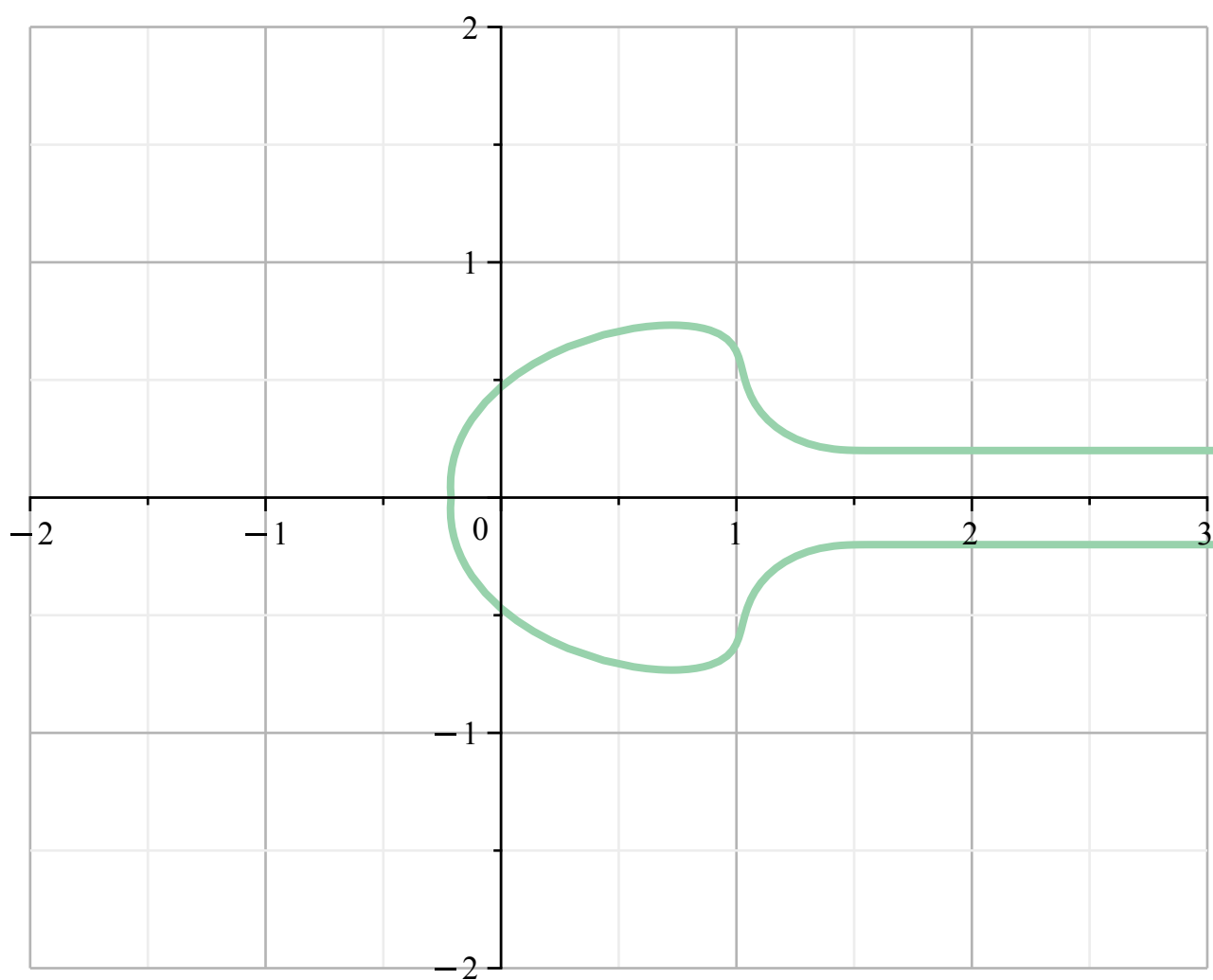
η_1 og η_2 danner sammen den ene halvdel af løkken i x- og y-retningen. For at opnå hele løkken, skal kurvedelen spejlvendt langs x-aksen. Dette opnåes ved at anvende både en positive og negative η_2 ;

```
> knotFig1:= display(plot([b*eta1(s), c*eta2(s)+a, s=-10..kx],
    color=mg, thickness=3, gridlines), plot([b*eta1(s), -c*eta2(s)-
    a, s=-10..kx], color=mg, gridlines, thickness=3), view=[-3...2,
    -2...2], scaling=constrained)
```



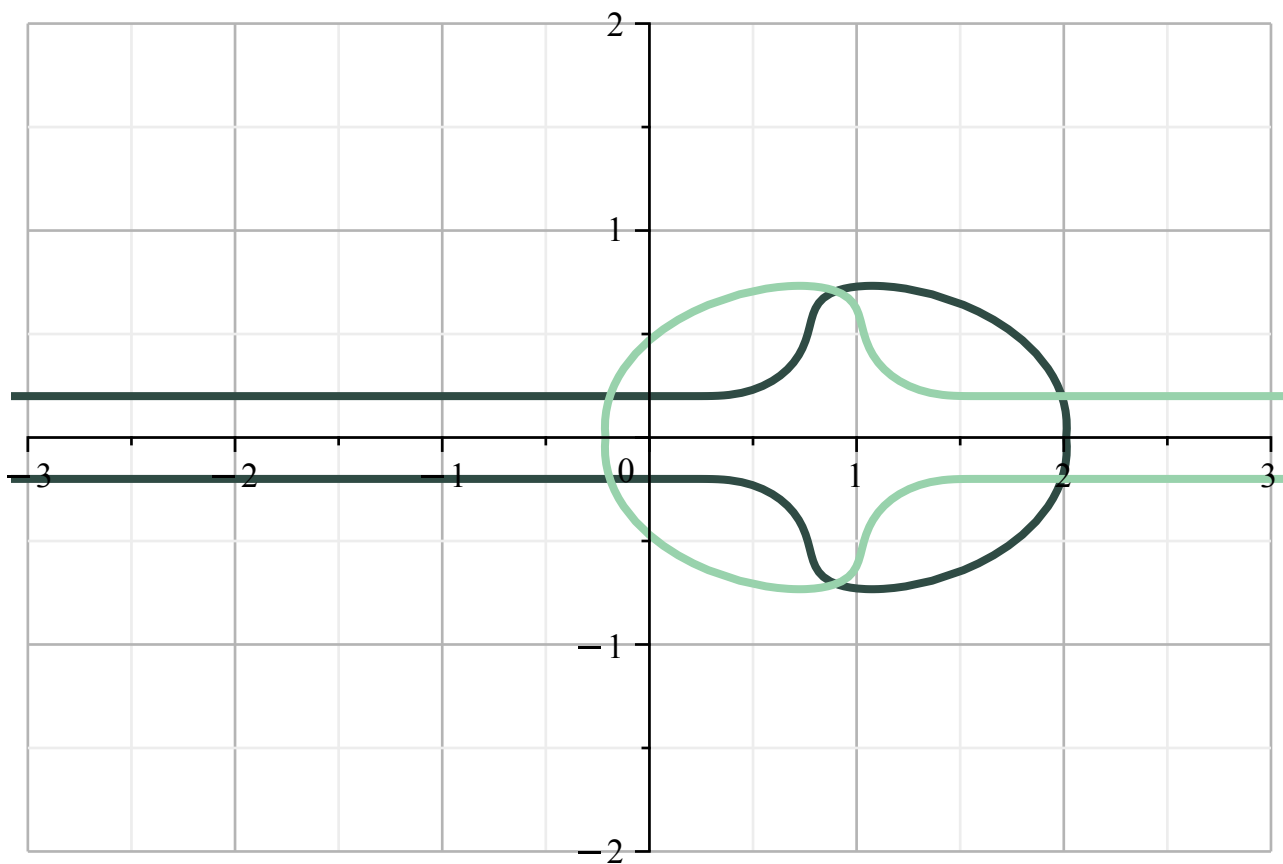
Herefter gentages konstruktionen for en spejlvendt version af den tidligere kurve:

```
> knotFig2:= display(plot([b*(d-eta1(s)), c*eta2(s)+a, s=-10..kx],
color=lg, gridlines, thickness=3), plot([b*(d-eta1(s)), -c*eta2
(s)-a, s=-10..kx], color=lg, gridlines, thickness=3), view=[-2..
.3, -2...2], scaling=constrained);
```



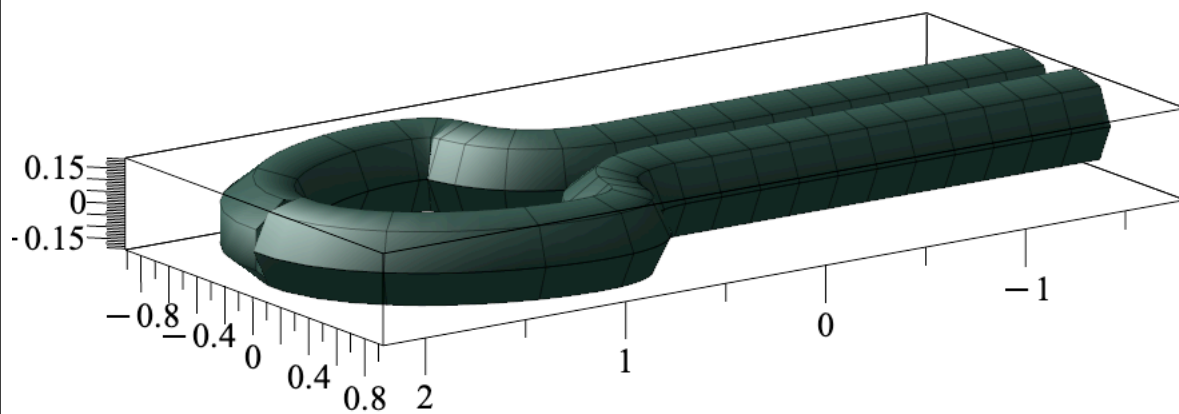
De to løkker samlet i xy-planet:

```
> display(knotFig1, knotFig2);
```



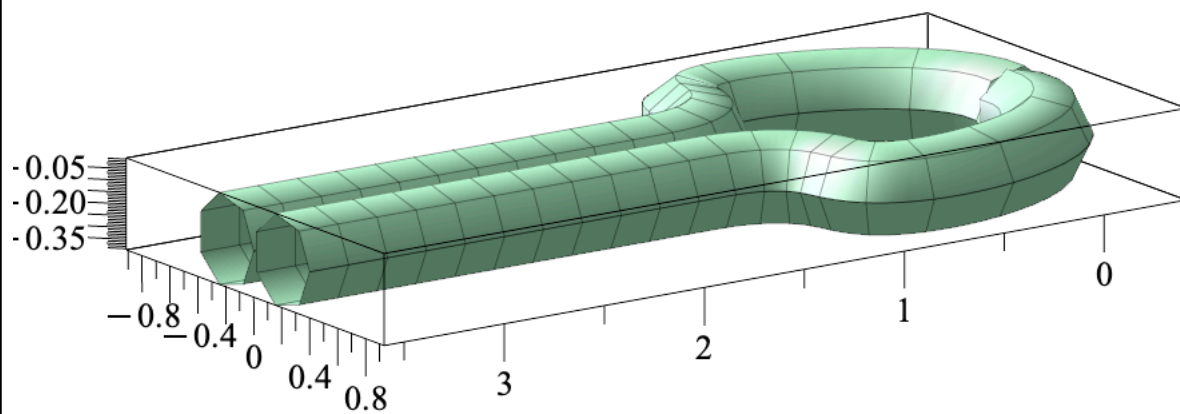
Knobet visualiseret i 3D:

```
> knot1B:= display(tubeplot([b*(eta1(s)), c*eta2(s)+a, 0], radius=
a, s=-2...kx, grid=[100,8], color=mg,scaling=constrained),
tubeplot([b*(eta1(s)), -c*eta2(s)-a, 0], radius=a, s=-2...kx,
grid=[100,8], color=mg,scaling=constrained));
```

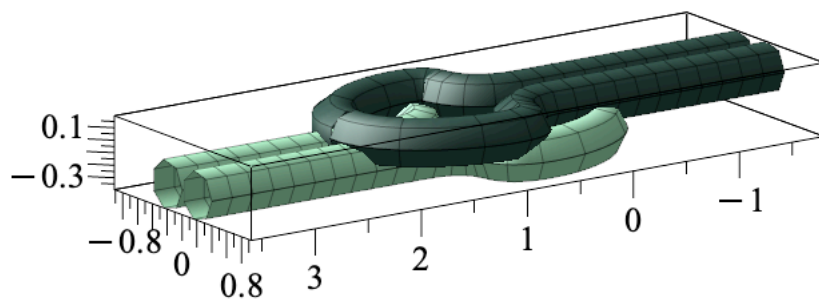


```
> knot2B:= display(tubeplot([b*(d-eta1(s)), c*eta2(s)+a, -a],
  radius=a, s=-2...kx, grid=[100,8], color=lg,scaling=
```

```
constrained),tubeplot([b*(d-eta1(s)), -c*eta2(s)-a, -a], radius=
a, s=-2...kx, grid=[100,8], color=lg,scaling=constrained));
```



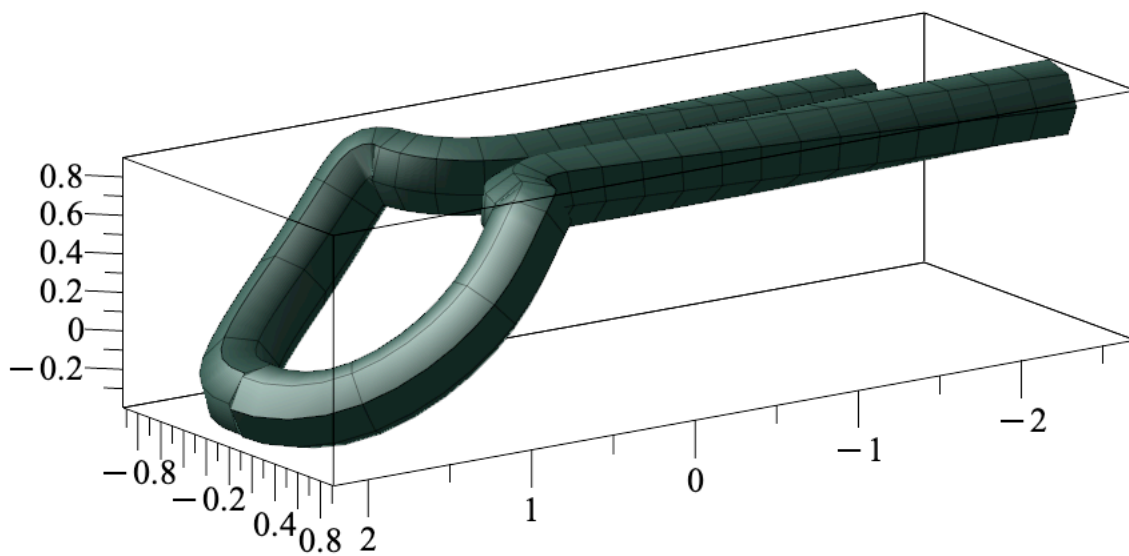
```
> display(knot1B, knot2B);
```



Dannelse af råbåndsknabet

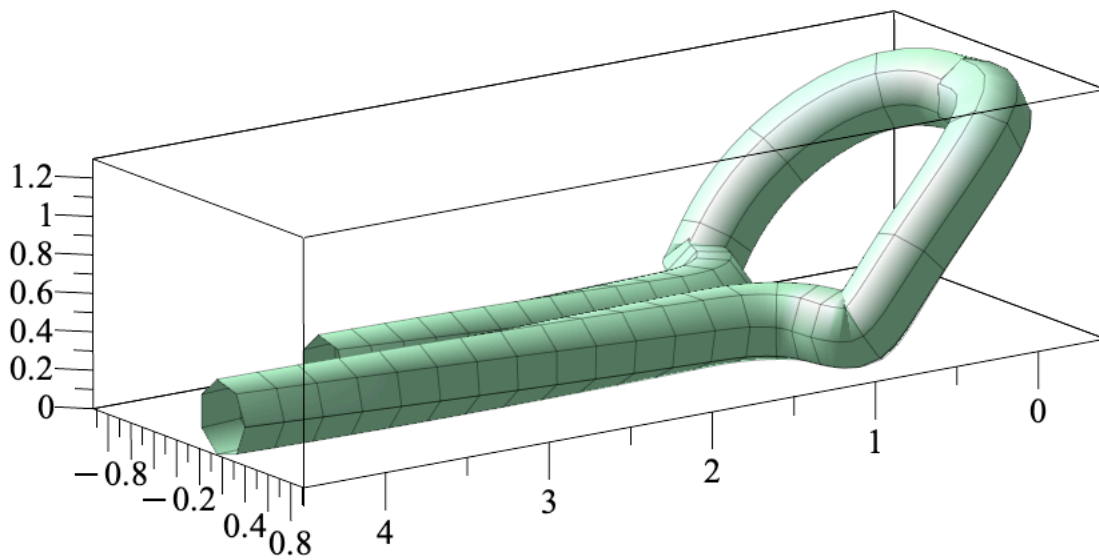
[Samling af xy-planet og z-kurven:

```
sq1:=display(tubeplot([(b*eta1(s)), c* eta2(s)+a, -a+e*g1(s)],
radius=a, s=-3...kx, grid=[100,8], color=mg,scaling=
constrained),
tubeplot([(b*eta1(s)), -c*eta2(s)-a, -a+e*g1(s)], radius=a, s=
-2...kx, grid=[100,8], color=mg,scaling=constrained));
```



```
> sq2:=display(tubeplot([b*(d-eta1(s)), c*eta2(s)+a, a+e*g2(s)],
  radius=a, s=-3...kx, grid=[100,8], color=lg,scaling=
```

```
constrained),tubeplot([b*(d-eta1(s)), -c*eta2(s)-a, a+e*g2(s)],  
radius=a, s=-2...kx, grid=[100,8], color=lg,scaling=  
constrained));
```



Begge dele af råbåndsknabet samlet:

```
> display(sq1,sq2, labels=["x","y","z"]);
```

